

Flyway CI/CD Pipeline with Jenkins

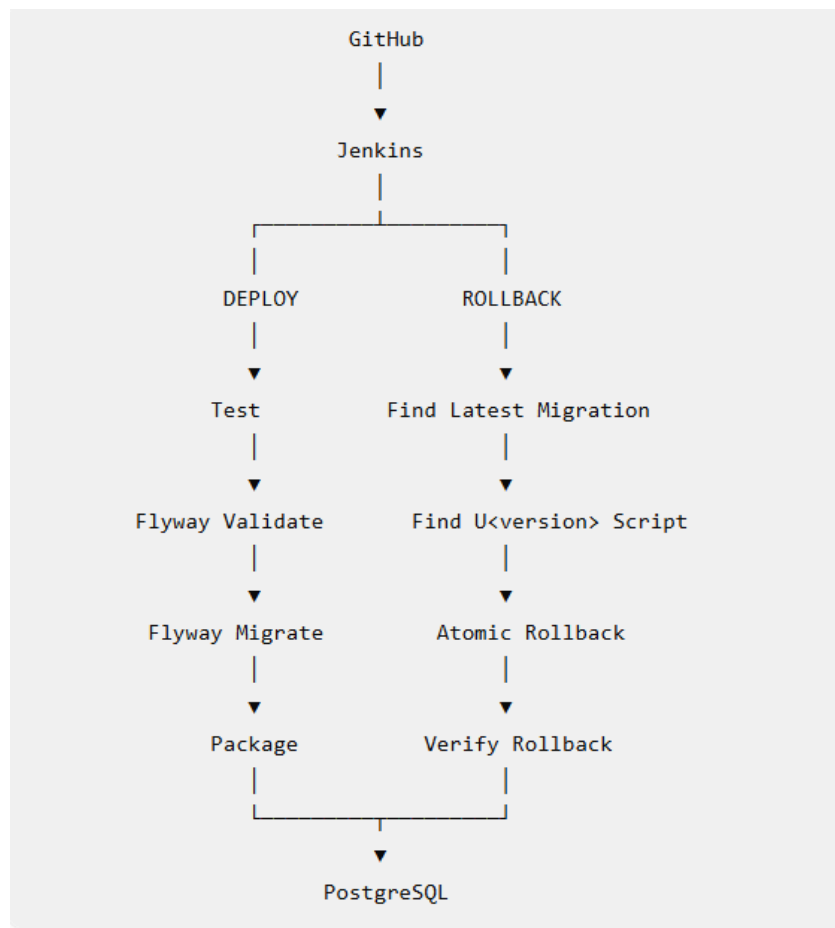
1. Overview

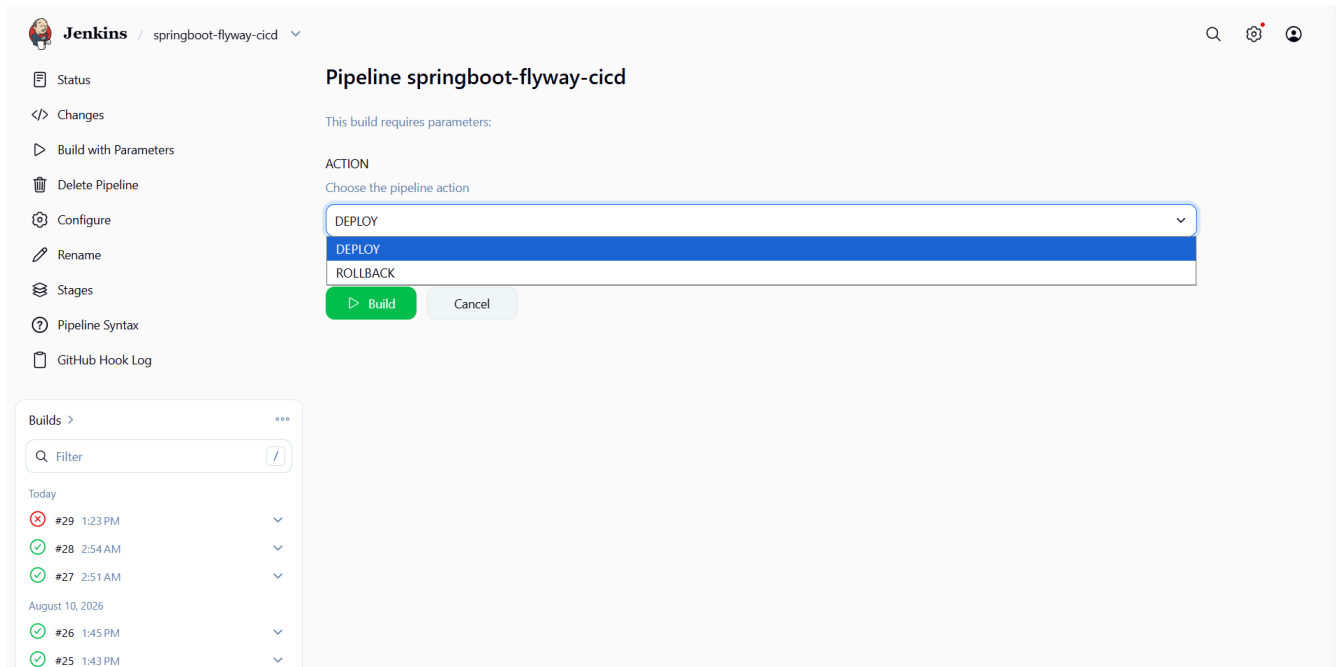
This project implements a CI/CD pipeline using **Jenkins**, **Spring Boot**, **PostgreSQL**, and **Flyway**.

The pipeline provides two actions:

- **DEPLOY** — runs tests, validates Flyway migrations, applies pending migrations, and packages the application.
- **ROLLBACK** — automatically identifies the latest successful migration, locates its corresponding rollback script, executes the rollback atomically, and verifies the result.

2. Pipeline Architecture





3. DEPLOY Pipeline

When **DEPLOY** is selected, Jenkins executes the following stages:

Test
↓
Flyway Validate
↓
Flyway Migrate
↓
Package

Test

The application tests are executed against the PostgreSQL database.

Flyway Validate

Flyway validates the migration files and checks the migration state of the database.

Flyway Migrate

Pending migrations are applied to the database.

Package

The Spring Boot application is packaged after successful testing and migration.

Case 1 — Successful Deployment and Migration

A successful deployment confirms that:

- Tests pass.
- Flyway validation succeeds.
- Pending migrations are applied.
- The application is packaged successfully.

Flyway Migration History Before Deployment

```
PS C:\Users\Maithreyi> docker exec jenkins bash -c "PGPASSWORD='devpass' psql -h host.docker.internal -p 5432 -U devuser -d flyway_dev -c 'SELECT installed_rank, version, description, success FROM flyway_schema_history ORDER BY installed_rank;'"
installed_rank | version | description          | success
-----+-----+-----+-----
1 | 1 | create customer      | t
2 | 2 | add customer kyc status | t
3 | 3 | add customer status   | t
4 | 4 | create customer account | t
(4 rows)
```

```
PS C:\Users\Maithreyi> |
```

Intermediate Stages execution in Jenkins Pipeline



Jenkins / springboot-flyway-cicd ▾ / Stages

Stages

August 11, 2026

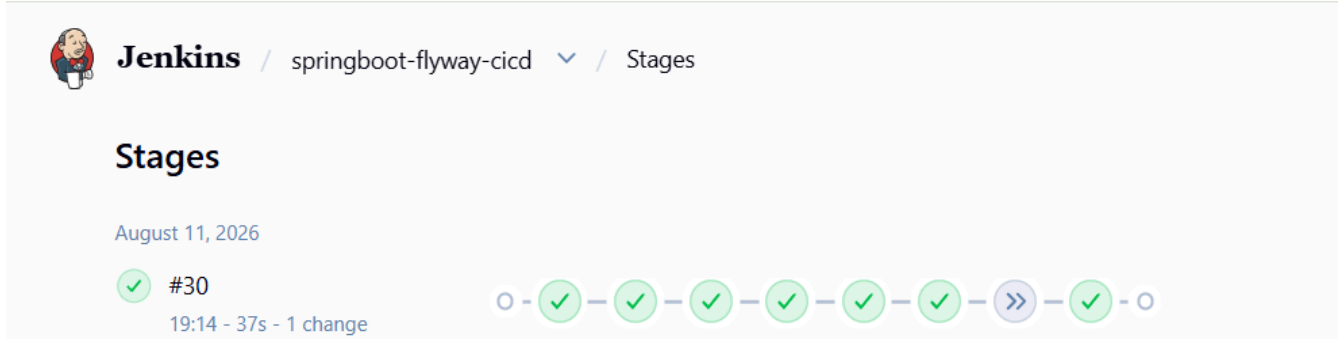


#30

19:14 - <1ms - 1 change



Successful DEPLOY pipeline



The image shows the Jenkins web interface for a pipeline named 'springboot-flyway-cicd'. The 'Stages' section is active, showing a successful build #30 on August 11, 2026. The build duration is 19:14 - 37s with 1 change. The pipeline progress bar consists of 10 stages, all marked with green checkmarks, indicating a successful deployment.

Flyway migration history after deployment

```
PS C:\Users\Maithreyi> docker exec jenkins bash -c "PGPASSWORD='devpass' psql -h host.docker.internal -p 5432 -U devuser -d flyway_dev -c 'SELECT installed_rank, version, description, success FROM flyway_schema_history ORDER BY installed_rank;'"
installed_rank | version | description          | success
-----+-----+-----+-----
1 | 1 | create customer      | t
2 | 2 | add customer kyc status | t
3 | 3 | add customer status   | t
4 | 4 | create customer account | t
5 | 5 | add account status     | t
(5 rows)
```

4. Automated Rollback

When **ROLLBACK** is selected, Jenkins does not receive a migration version from the user.

Instead, it determines the migration automatically.

Rollback Process

Read flyway_schema_history



Find latest successful migration



Extract version



Find U<version>__*.sql



Validate exactly one script exists



Execute rollback in transaction

↓
Delete corresponding history entry
↓
Verify rollback

For example, if the latest migration is **V5**, Jenkins automatically searches for:

U5__*.sql

5. Rollback Validation

Before executing the rollback, Jenkins validates the rollback script.

Validation rules

Condition	Result
No matching rollback script	Rollback fails
Multiple matching scripts	Rollback fails
Exactly one matching script	Rollback proceeds

This prevents Jenkins from executing an ambiguous or missing rollback script.

Case 2 — Failed Rollback: Missing Rollback Script

If the latest migration does not have a corresponding rollback script, the pipeline stops before making database changes.

Expected behavior:

ERROR: No rollback script found for version X.

Failed Rollback Script



6. Atomic Rollback

Once validation succeeds, the rollback is executed inside a PostgreSQL transaction.

```
BEGIN;
```

```
-- Execute rollback SQL
```

```
-- Remove corresponding Flyway history entry
```

```
-- Validate deletion
```

```
COMMIT;
```

The rollback therefore treats:

- Database schema reversal
- Flyway history removal

as a single atomic operation.

If the transaction fails, the database changes are not committed.

7. Case 3 — Successful Rollback

For the successful V5 rollback:

Before rollback

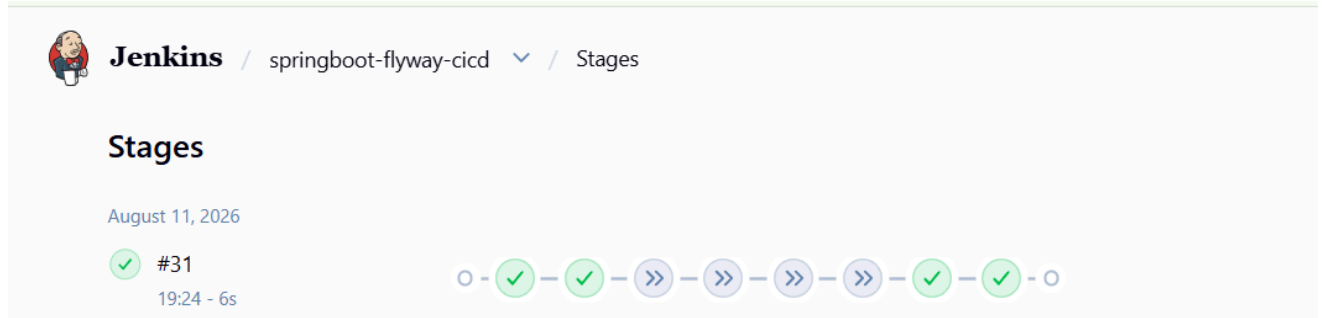
V1 → V2 → V3 → V4 → V5

The pipeline automatically detected:

Latest migration : V5__add_account_status.sql

Rollback script : U5__rollback_account_status.sql

Successful rollback



The image shows the Jenkins web interface for a pipeline named 'springboot-flyway-cicd'. The 'Stages' section displays the execution history for build #31, dated August 11, 2026, at 19:24, which took 6 seconds to complete. The pipeline consists of 10 stages, represented by a sequence of icons: a green circle with a checkmark, followed by two green circles with checkmarks, then four blue circles with double right-pointing arrows, and finally two green circles with checkmarks, ending with a small open circle. This indicates a successful completion of the pipeline.

8. Rollback Verification

The rollback is verified at two levels.

Flyway History

After rollback:

V1 → V2 → V3 → V4

V5 must no longer exist in `flyway_schema_history`.

Flyway history after successful rollback

```
PS C:\Users\Maithreyi> docker exec jenkins bash -c "PGPASSWORD='devpass' psql -h host.docker.internal -p 5432 -U devuser -d flyway_dev -c 'SELECT installed_rank, version, description, success FROM flyway_schema_history ORDER BY installed_rank;'"
installed_rank | version | description          | success
-----+-----+-----+-----
1 | 1 | create customer      | t
2 | 2 | add customer kyc status | t
3 | 3 | add customer status   | t
4 | 4 | create customer account | t
(4 rows)
```

Database Schema

Before Rollback:

```
PS C:\Users\Maithreyi> docker exec jenkins bash -c "PGPASSWORD='devpass' psql -h host.docker.internal -p 5432 -U devuser -d flyway_dev -c '\d customer_account'"
Table "public.customer_account"
  Column      |          Type          | Collation | Nullable |          Default          |
--------------+-----+-----+-----+-----+-----+
 account_id   | bigint                 |           | not null | nextval('customer_account_account_id_seq'::regclass) |
 customer_id  | bigint                 |           | not null |                               |
 account_number | character varying(20) |           | not null |                               |
 account_type | character varying(20) |           | not null |                               |
 balance      | numeric(15,2)          |           | not null | 0.00                      |
 created_at   | timestamp without time zone |           | not null | CURRENT_TIMESTAMP        |
 account_status | character varying(20) |           | not null | 'ACTIVE'::character varying |
Indexes:
    "customer_account_pkey" PRIMARY KEY, btree (account_id)
    "customer_account_account_number_key" UNIQUE CONSTRAINT, btree (account_number)
Foreign-key constraints:
    "fk_customer_account_customer" FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
```

For the V5 test, the `account_status` column previously introduced by V5 is no longer present in `customer_account`.

After Rollback:

```
PS C:\Users\Maithreyi> docker exec jenkins bash -c "PGPASSWORD='devpass' psql -h host.docker.internal -p 5432 -U devuser -d flyway_dev -c '\d customer_account'"
Table "public.customer_account"
  Column      |          Type          | Collation | Nullable |          Default          |
--------------+-----+-----+-----+-----+
 account_id   | bigint                 |           | not null | nextval('customer_account_account_id_seq'::regclass) |
 customer_id  | bigint                 |           | not null |                               |
 account_number | character varying(20) |           | not null |                               |
 account_type | character varying(20) |           | not null |                               |
 balance      | numeric(15,2)          |           | not null | 0.00                      |
 created_at   | timestamp without time zone |           | not null | CURRENT_TIMESTAMP        |
Indexes:
    "customer_account_pkey" PRIMARY KEY, btree (account_id)
    "customer_account_account_number_key" UNIQUE CONSTRAINT, btree (account_number)
Foreign-key constraints:
    "fk_customer_account_customer" FOREIGN KEY (customer_id) REFERENCES customer(customer_id)
```